Engine Model(s): PT6A

79-20-02

OIL FILTER, HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire used shall, unless otherwise specified, comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name	Remarks
PWC05-019	Compound, Lapping	
PWC05-064	Solution, Anodize Touch-up	
PWC05-145	(Use PWC05-019)	
PWC11-014	Alcohol, Isopropyl	
PWC11-027	Solvent, Petroleum	
PWC11-031	Cleaner, Engine	
PWC11-074	Solvent, Non-Aqueous	
PWC11-084	Degreaser, Water-Repellent	

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30328	Puller, Filter Housing	(Obsolete - Replaced by PWC50726)
PWC30556	Puller	
PWC50726	Knocker, Oil Filter Housing	(Alternative to PWC30328)

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

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Name	Remarks
Cleaner, Electrosonic	
Vacuum Pump	Oil Filter Element Rinsing and Debris Apparatus Kit

5. Servicing

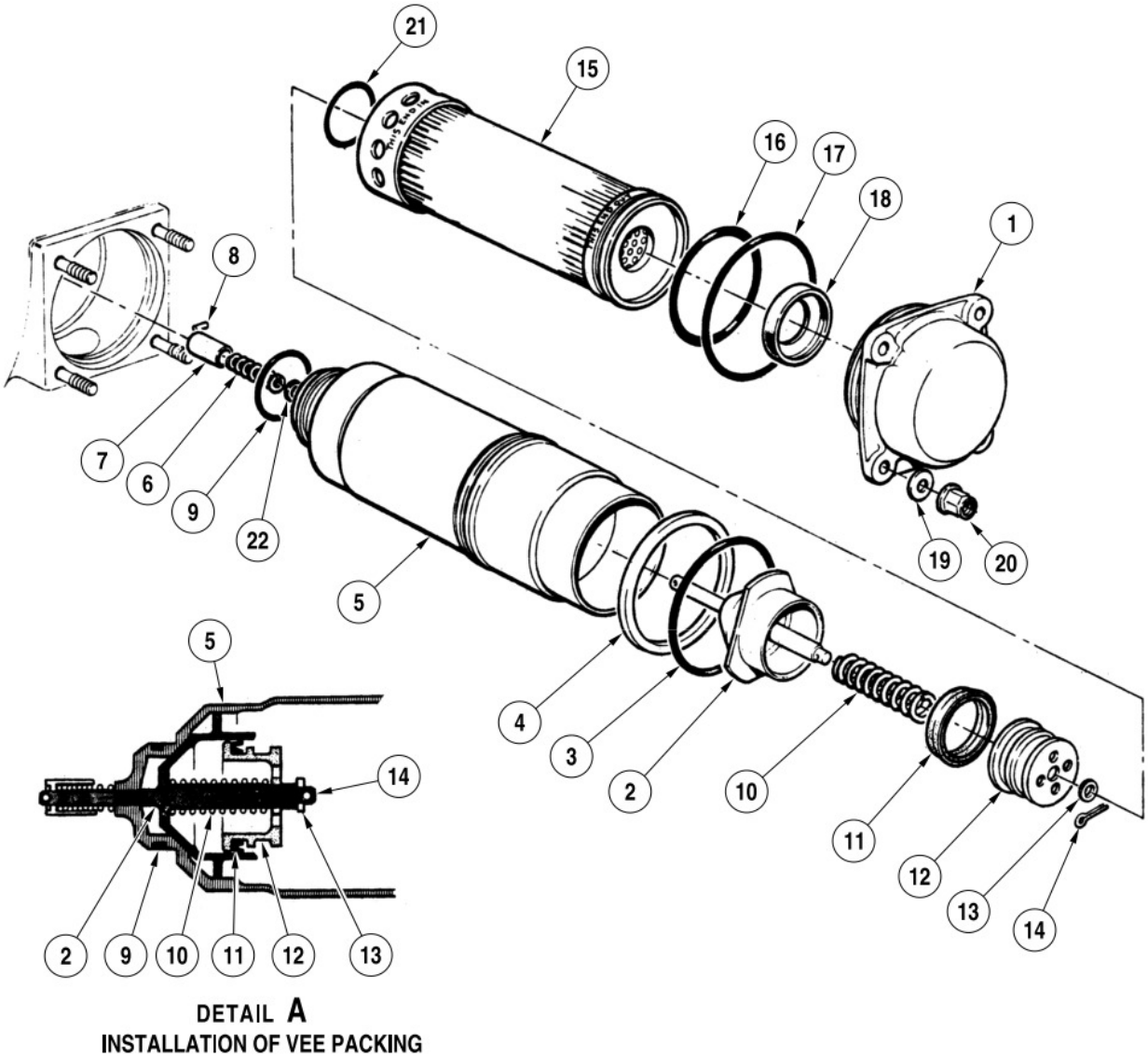
For oil system servicing and inspection criteria, refer to Chapter 72-00-00, SERVICING and INSPECTION, respectively.

6. Removal/Installation

A. Removal of Oil Filter Element (Ref. Fig. 201 and 202)

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Figure 201 Removal/Installation of Oil Filter Element, Housing and Bypass/Check Valve



Pre-SB1247

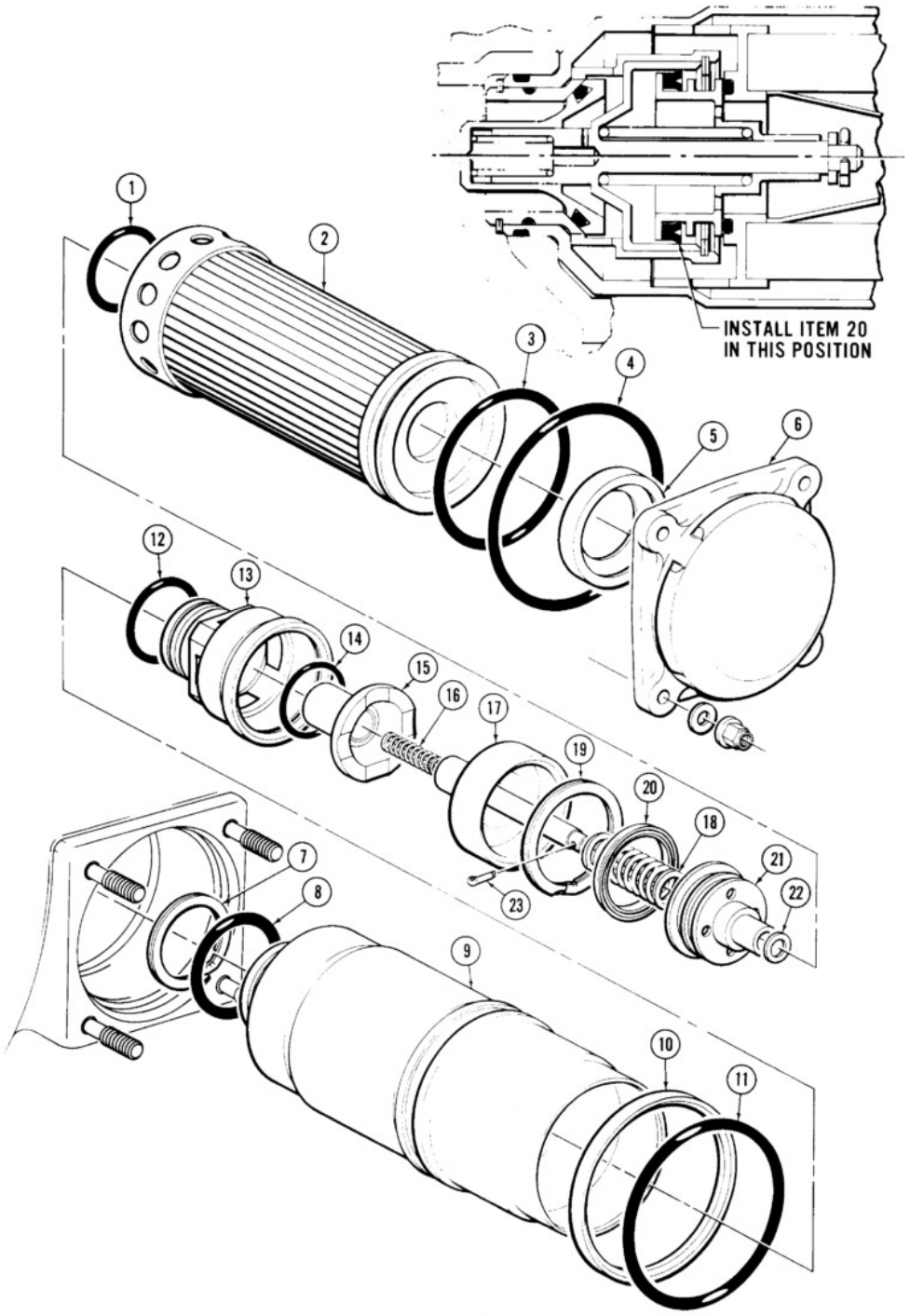
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1. Oil Filter Cover
2. Check Valve
3. Preformed Packing
4. Plastic Ring
5. Oil Filter Housing
6. Check Valve Spring
7. Spring Retainer Collar
8. Lockpin
9. Preformed Packing
10. Bypass Valve Spring
11. Packing
12. Bypass Valve Piston
13. Washer
14. Cotterpin
15. Oil Filter Element
16. Preformed Packing
17. Preformed Packing
18. Teflon Spacer
19. Washer
20. Nut
21. Preformed Packing
22. Repair Washer (if fitted)

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Figure 202 Removal/Installation of Oil Filter Element, Housing and Bypass/Check Valve - Post-SB1247



Post-SB1247 and Post-SB1379

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1. Preformed Packing
2. Check Valve
3. Preformed Packing
4. Preformed Packing
5. Teflon Spacer
6. Oil Filter Cover
7. Retaining Ring
8. Preformed Packing
9. Oil Filter Housing
10. Plastic Ring
11. Preformed Packing
12. Preformed Packing
13. Valve Housing
14. Preformed Packing Engines)
15. Check Valve
16. Check Valve Spring
17. Bypass Valve Guide
18. Spring
19. Retaining Ring
20. Preformed Packing (V-type)
21. Bypass Valve
22. Washer
23. Cotterpin

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(1) Remove self-locking nuts and plain washers fastening oil filter cover to compressor inlet case.

(2) Remove cover from case, complete with preformed packing and teflon spacer in-situ. Discard preformed packing.

NOTE: Do not remove teflon spacer from cover unless damaged.

(3) Withdraw filter element from oil filter housing using puller (PWC30556):

(a) Unscrew knurled nut to end of thread on rod.

(b) Screw rod into collet assembly until enlarged end of rod clears collet teeth.

(c) Install puller over filter housing, so the end of the housing seats on the counterbore of puller.

(d) Push rod inward until collet teeth engage behind inner bore of filter element end cover.

(e) Carefully unscrew rod to lock collet teeth in position.

NOTE: Do not over expand collet teeth, as teeth may contact and damage center support tube of filter element.

(f) Screw in knurled nut to withdraw filter element.

(g) Remove puller and element from filter housing, then remove puller from element.

(4) Discard preformed packings from element.

B. Removal of Oil Filter Housing (Ref. Fig. 203)

(1) Drain engine oil (Ref. 72-00-00, SERVICING).

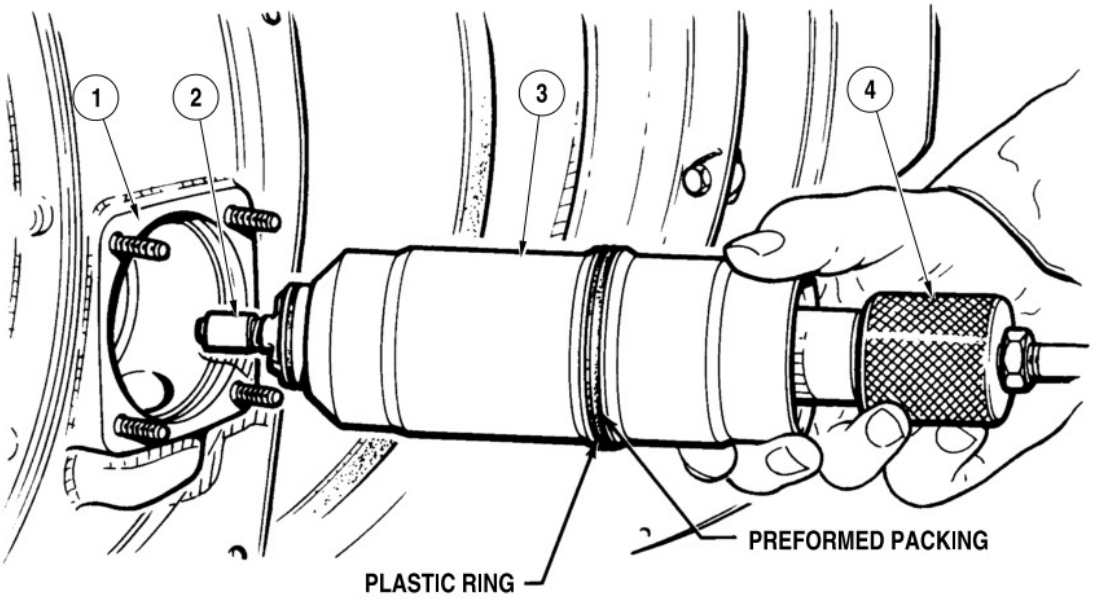
(2) Remove oil filter element (Ref. Para. A.).

(3) Remove the oil filter housing assembly (3), from the compressor inlet case with puller (PWC30328) or (PWC50726).

(4) Remove preformed packings and plastic ring from housing.

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Figure 203 Removal/Installation of Oil Filter Housing - Typical



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1. Oil Filter Cover Mount Pad
2. Check Valve Collar
3. Oil Filter Housing
4. Oil Filter Housing Puller

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C. Removal of Oil Filter Check/Bypass Valve Assembly (Pre-SB1247) (Ref. Fig. 201)

(1) With housing help vertically (check the valve spring (6) downward), compress the spring and remove the lockpin (8) and spring retainer collar (7), spring (6) and repair washer (22), if fitted.

NOTE: If repair washer (11) is fitted and housing is serviceable, tag the housing as a repaired item and retain the washer for reinstallation. If the housing is to be replaced, discard the repair washer.

(2) Invert the housing to facilitate the check and bypass valve assembly (2 and 10 through 14) removal. Care must be exercised to prevent damage to the valve seat.

(3) If a new check valve and/or bypass valve is to be installed or check valve seating requires lapping (Ref. Para. 9.A. and B.), disassemble the check and bypass valve assembly as follows:

(a) Remove the cotterpin (14), washer (13), bypass valve piston (12), packing (11) and bypass valve spring (10) from the check valve (2).

D. Removal of Oil Filter Check/Bypass Valve Assembly (Post-SB1247) and (Post-SB1379) (Ref. Fig. 202)

(1) Drain engine oil (Ref. 72-00-00, SERVICING).

(2) Remove oil filter element (Ref. Para. A.).

(3) Exercising care, remove the retaining ring (7) and withraw the check/bypass valve assembly from the filter housing (9).

(4) Remove the preformed packing (12) from the check valve seat (13).

(5) Remove the retaining ring (19) and separate the bypass valve guide (17) from the valve seat (13). Exercise care when removing the check valve (15) as it is spring loaded to the valve guide.

(6) Remove the check valve (15) and spring (16) from the valve seat.

(7) Remove the preformed packing (14) if applicable, from the valve (15).

NOTE: Check valves for Post-SB1379 engines differ from the previous check valves, in that the preformed packing has been deleted.

CAUTION: EXERCISE CARE DURING THIS OPERATION, AS PARTS ARE SPRING LOADED.

(8) Apply light pressure to the bypass valve (21) and remove the cotterpin (23) and washer (22). Separate the assembly.

(9) Remove the spring (18) from the shaft of the guide and the preformed packing from the valve (21).

E. Installation of Oil Filter Element (Ref. Figs. 201 and 202)

(1) Install new preformed packings on the filter element (3).

(2) Install filter element, perforated flanged end first, into the filter housing in the compressor

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inlet case.

NOTE: The ends of the filter element Post-SB1215 are marked respectively "THIS END IN" and "THIS END OUT".

(3) Install a new lubricated preformed packing on the filter cover.

NOTE: Make sure the teflon spacer is in position on the lugs of the filter cover.

(4) Install the filter cover on the compressor inlet case with washers and self-locking nuts. Tighten nuts 32 to 36 lb.in.

(5) Start engine and do checks associated with Installation/Replacement of Oil Filter, Housing, and Check Valve (Ref. 71-00-00, ADJUSTMENT/TEST, Table 501).

F. Installation of Oil Filter Housing (Ref. Figs. 201 and Fig. 202)

(1) Install a new preformed packing in the groove at the inner end of the oil filter housing.

(2) Install a plastic ring in the inner section of the groove at the center of the housing.

(3) Install a preformed packing in the outer section of the groove and butt against the plastic ring.

(4) Insert the filter housing assembly into the compressor inlet case and, using hand pressure, push the assembly until firmly seated.

NOTE: Do not apply side loads on the housing while it is being pressed into the inlet case.

(5) Install the oil filter element (Ref. Para. E.).

(6) Fill the oil tank (Ref. 72-00-00, SERVICING).

(7) Start the engine and do the checks associated with Installation/Replacement of Oil Filter, Housing, and Check Valve (Ref. 71-00-00, ADJUSTMENT/TEST, Table 501).

G. Installation of Oil Filter Check/Bypass Valve Assembly (Pre-SB1247) (Ref. Fig. 201)

(1) If a new check value and/or bypass valve is to be installed or if the check valve seat has been re-lapped, proceed as follows:

(a) Install the bypass valve spring (10) over the stem on the check valve (2).

(b) Install the packing (11) in the inner groove of the bypass valve piston (12), with the open end of the packing facing the outer groove of the piston (Detail A).

(c) Install the bypass valve piston over the spring (10) and into the centerbore of the check valve (2).

(d) Compress the spring until the stem protrudes from the end of the piston and secure the piston on the stem with a washer (13) and cotterpin (14).

(2) Make sure the inside of the oil filter housing is clean of all residue.

(3) Slide the check and bypass valve assembly into the filter housing until seated.

(4) Hold the valve assembly in position with a suitable fiber drift and install the check valve spring (6) and spring retaining collar (7) on the protruding section of the check valve stem.

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NOTE: If a repaired filter housing (Ref. Para. C.) is being reinstalled, make sure the repair washer (22) is installed between the shoulder on the housing and check valve spring.

(5) Compress thee spring with the collar and insert the lockpin (8) through the hole in the valve stem. Release the spring and allow the collar to slip over the lockpin. Make sure the lockpin is completely covered by the retaining collar.

(6) Carry out a static leak test (Ref. Para. 9.D.).

H. Installation of Oil Filter Check/Bypass Valve Assembly (Post-SB1247) and (Post-SB1379) (Ref. Fig. 202)

(1) Install the V-type preformed packing (10) on the bypass valve (21). Make sure the packing is installed with the open end of the V facing the outwards as shown.

(2) Install the spring (18) on the internal shaft of the bypass valve guide (17). Locate the bypass valve (21) over the shaft and compress the spring to that the shaft protrudes from the end of the valve. Secure the assembly with the washer (22) and cotterpin (23).

(3) Install the Pre-SB1379 preformed packing (14) on the seat of the check valve (15).

(4) To improve the oil filter housing check valve sealing, the Pre-SB1379 check valve is replaced with a new Post-SB1379 check valve having no preformed packing groove. Prior to installation, a new check valve and/or seat must be lapped (Ref. Para. 9.B.).

(5) Install spring (16) in the recess in the external shaft of the bypass valve guide (17). Install the complete assembly in the valve seat (13) and secure with a retaining ring (19).

(6) Install a preformed packing (12) on the valve seat (13).

CAUTION: PRIOR TO INSERTION OF THE VALVE ASSEMBLY INTO THE HOUSING, MAKE SURE THE PREFORMED PACKING (12) IS CORRECTLY SEATED ON THE VALVE SEAT AND IS NOT DISTORTED.

(7) Make sure the inside of the oil filter housing is clean of all residue.

(8) Install the complete assembly of the check/bypass valve in the oil filter housing and secure with retaining ring (7).

(9) Do a static leak test (Ref. Para. 9.D.).

7. Cleaning

A. General

For periodic cleaning requirements, refer to Chapter 72-00-00.

B. Cleaning of Oil Filter Element (Pre-SB1215)

(1) If this method of cleaning the filter element, or the method detailed in Para. D. proves to be ineffective, Pre-SB1215 elements may be ultrasonically cleaned at an overhaul facility.

(a) Seal central tube opening at both ends of oil filter element with suitable sized rubber plugs.

(b) Agitate element for five minutes in clean, unused solvent (PWC11-027) or (PWC11-031).

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(c) Dry element with clean, dry compressed air or allow the element to stand in a clean environment until dry.

(d) Inspect element (Ref. Inspection/Check).

(e) Repeat cleaning and inspection, if required, until element is considered serviceable.

NOTE: Use clean, unused solvent for each repeated cleaning operation.

C. Cleaning of Oil Filter Element (Post-SB1215 and Post-SB1282)

CAUTION: DO NOT ULTRASONICALLY CLEAN, PRESSURE FLUSH OR DRY ELEMENT WITH COMPRESSED AIR. SUCH CLEANING AND DRYING WOULD CAUSE DAMAGE TO THE FILTER MEDIA.

NOTE: If this method of cleaning filter element proves ineffective, the alternate method detailed in Para. D. may be used.

(1) Gently agitate the element in unused petroleum solvent (PWC11-027) or (PWC11-031).

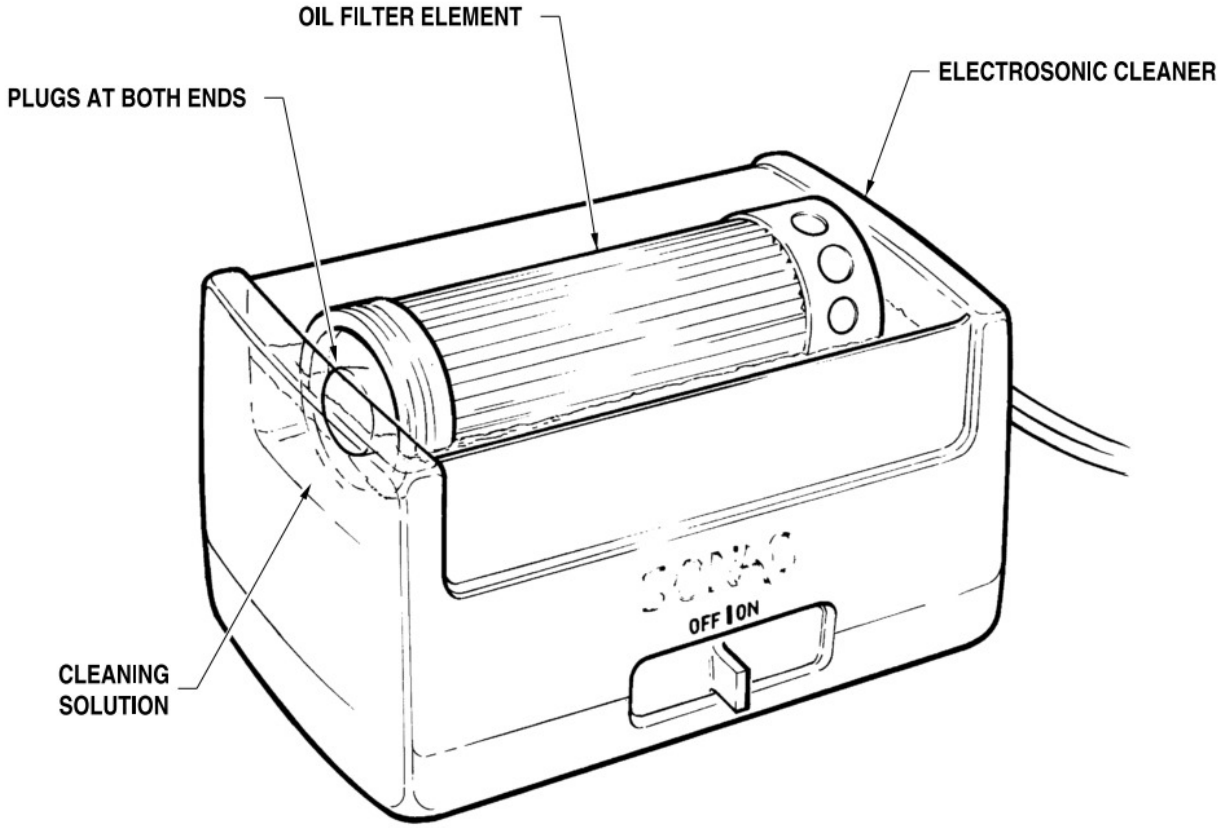
(2) Allow the element to stand in a clean environment until dry.

(3) Inspect element (Ref. Inspection/Check).

D. Electrosonic Wash (Ref. Fig. 204)

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Figure 204 Electrosonic Cleaner - Oil Filter Element



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(1) Plug openings at each end of filter element with suitable plastic or rubber plugs.

(2) Place element in cleaning tank of electrosonic cleaner.

(3) Put sufficient quantity of isopropyl alcohol (PWC11-014) in tank to a level approximately 1/4 inch below top.

(4) Operate unit for 10 minutes.

(5) Rotate element 180 degrees (end-to-end) in tank and reoperate unit for 10 minutes.

NOTE: Cleaning duration should be adjusted depending on condition of element. However, for optimum results, cleaning solvent should be replaced every 20 minutes.

(6) Using clean, lint-free gloves, remove element from tank and remove plugs.

(7) Allow element to stand in clean environment until dry.

(8) Vibropeen cleaning date on end cap.

(9) Inspect element (Ref. Inspection/Check).

8. Inspection/Check

A. Main Oil Filter Inspection

(1) Put a small lamp into the inner bore of filter element.

(2) Use a magnifying glass and visually examine the oil filter and secondary screen (coarse hat-type screen attached to the inner end of the filter) for breaks, holes, tears or signs of crushing or collapse. There must be no loose particles, broken bonds or joints.

NOTE: Not applicable on main oil filter without the secondary screen configuration.

(3) If the external surface of the oil filter is damaged, then replace the main oil filter.

(4) If there is damage or opening on the secondary screen, then do an inspection (Ref. 72-00-00, ENGINE, TURBOPROP - INSPECTION).

B. Main Oil Filter Contamination and Chip Detector Circuit Completion

(1) After an engine/module change due to bearing, gear or bronze bushing distress, it is recommended that the replacement engine have an oil filter patch-check (Ref. Para. 8.C.) after 50 hours. This is to make sure the debris from the original failure has not contaminated the replacement engine, due to incomplete flushing of the airframe/propeller oil system(s).

(2) If magnetic or bronze material is found, as a contaminant, repeat patch check after 50 hours. If the amount of debris has reduced, repeat patch check at 50 hour intervals until no debris (traces are acceptable, except bearing material) is found.

(3) The time between patch checks should revert to standard maintenance intervals after the second consecutive clean patch check. If the amount of magnetic/bronze material increases or if magnetic/bronze material is found, as a major or minor contaminant, after the second clean patch check, the maintenance action required for debris in oil system must be carried out.

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(4) In all instances when engine generated metal contamination requiring engine removal occurs and the main oil filter is bypassed, the airframe oil cooler must be replaced, (Ref. Aircraft Maintenance Manual) and airframe oil tubes and hoses flushed. Failure to do so will cause contamination of the replacement engine.

(5) Chip detector circuit completion with main oil filter bypass requires engine removal.

NOTE: An assessment of the amount of contamination can indicate whether a main oil filter bypass has actually occurred. If the main oil filter is cleaned or replaced at regular intervals as recommended by P&WC and debris is deposited on most of the surface of filter element and associated housing, it can be considered that a bypass has occurred.

(6) Chip detector circuit completion not with main oil filter bypass requires removal and inspection of the chip detector and associated strainers and main oil filter. Do patch-making procedure on main oil filter and transfer any debris from chip detector (Ref. Subpara. 8.C.).

(7) Inspect chip detector and/or main oil filter debris using a magnifying glass and/or microscope.

(8) After inspection is complete, all debris collected from chip detector and oil filter may be sent to a laboratory for analysis (Ref. 70-00-00, MAINTENANCE PRACTICES).

(9) Record engine TSN, cycles, category, type and origin of debris for trend monitoring purposes.

(10) An operator has the option of doing the filter patch-check or sending the filter to a laboratory for analysis. However, an inspection of the collected debris must be done as quickly as possible to determine the required maintenance action. Keeping a debris generating engine/module is service while removal is planned may substantially increases the cost of repair/refurbishment due to further debris damage.

(11) Allowable Debris

(a) Small quantity of hair-like filaments (fuzz) or powder (including bronze particles), spiral curls (machining swarf) and silver plating.

(b) If silver plating is found, proceed as follows:

1 If the main oil filter contains a large quantity (15 or more) of silver flakes, the engine oil system must be flushed (Ref. Chapter 72-00-00 SERVICING).

2 If silver flakes greater than 0.080 inch are found in the main oil filter, the AGB internal scavenge pump screen must be inspected (Ref. Chapter 72-60-00, INSPECTION and CLEANING). The engine oil system must be flushed (Ref. Chapter 72-00-00 SERVICING).

3 In both cases after performing the above, the engine may be returned to service, and the oil filter reinspected at the next scheduled inspection (Ref. Chapter 72-00-00, INSPECTION).

(c) Collect allowable debris, clean and reinstall main oil filter, strainer, chip detector strainer(s) and chip detector. The engine may be returned to normal operation and oil

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filter and chip detector inspection to periodic inspection intervals.

(12) Non-allowable Debris

(a) Category 1:

1 (Identifiable fragments i.e., keywasher keys, tooth segments, etc.). If non-allowable Category 1 debris is found, replace affected module.

(b) Category 2:

1 Large quantities of hair-like filaments (fuzz) or powder (including bronze particles).

NOTE: Large quantity is defined as sufficient fuzz and/or powder filaments to completely cover the chip detector magnetic poles or more than 40 particles on a filter.

(c) Category 3:

1 Dark irregular magnetic chips (minimum dimension 0.010 in.).

2 Small clusters of magnetic flakes (three or more).

NOTE: Thin, shiny rolled flakes with feathered edges and more than 0.020 in. in size are generated when bearing surfaces break down due to excessive load (spalling). The outer surface of the flakes is highly polished and may show parallel impressions. The inner surface has a rough wavy or granular texture. After the bearing surface breaks down, the underlying material disintegrates and chips with dark, coarse and irregular shapes are produced.

3 All other debris not defined above.

(13) If non-allowable Category 2 or 3 debris is found and engine/module has no history of generating debris (within 400 hours), proceed as follows:

(a) Clean and install chip detectors, strainers and main oil filter (Ref. Removal/Installation).

(b) Start engine (Ref. 71-00-00, ADJUSTMENT/TEST) and run at 80% T.O. power for 10 minutes .

(c) Remove and inspect chip detector, associated strainers and main oil filter (Ref. Removal/Installation).

(d) If no debris is found, clean and reinstall chip detector, strainer and main oil filter.

(e) Return engine to service. Check chip detector and main oil filter after 10 hours, and providing no debris is found, after 50 hours. If no debris is found at 50 hours, return oil filter and chip detector inspection to periodic inspection intervals.

(f) If debris is found, flush engine (Ref. 72-00-00, SERVICING) and powerplant oil system (Ref. Aircraft Maintenance Manual). Install clean chip detector, strainers and main oil filter (Ref. Removal/Installation).

(g) Fill engine (Ref. 72-00-00, SERVICING) and powerplant oil system (Ref. Aircraft Maintenance Manual).

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(h) Start engine (Ref. 71-00-00, ADJUSTMENT/TEST) and run at 80% T.O. power for 10 minutes.

(i) Remove and inspect chip detector, strainers and main oil filter.

NOTE: It is possible that debris originating from power section module is generated by the propeller governor and/or overspeed governor (O/S governor). To determine the actual source of the debris, remove these components individually. Determine the source of debris within 50 hours.

(j) If no debris is found, install chip detector, strainers and main oil filter. Check chip detector and main oil filter after 10 hours and providing no debris is found, after 50 hours. If no debris is found at 50 hours, return to standard periodic inspection interval.

(k) If category 2 or 3 non-allowable debris is found, replace module/engine.

(14) If non-allowable Category 2 (not bronze) or Category 3 debris is found and the engine has a history of generating debris (within 400 hours), proceed as follows:

(a) Review the results of last laboratory analysis.

(b) If results showed bearing material and origin was from the engine and not the propeller interface unit or permanent magnetic alternator, then engine should be removed.

(c) If results were not bearing material and quantity of debris is not increasing or engine is not consistently generating debris, install chip detector, strainers and main oil filter. Return engine to service. Check chip detector and main oil filter after approximately 10 hours and providing no debris is found, after 50 hours. If no debris is found at 50 hours, return oil filter and chip detector inspection to standard periodic inspection intervals.

NOTE: Results and origin of debris should be determined within 50 flight hours of original detection of debris.

(d) If results were not bearing material or results of last sample are still unknown, and quantity of debris is same or increasing and engine is consistently generating debris, replace module/engine at first opportunity within 10 hours of current report of debris.

(15) If non-allowable category 2 (bronze) debris is found and the RGB has a history of generating similar bronze debris (within 400 hours), proceed as follows:

(a) If the RGB/power section is confirmed to be a replacement for one removed (within 100 hours) due to bronze bushing distress, clean and reinstall, main oil filter and strainers. Return the engine to service. Check chip detector and main oil filter after 10 hours and providing no debris is found, 50 hours. If no debris is found at 50 hours, return oil filter and chip detector to standard periodic inspection intervals.

(b) If the RGB/power section is not a replacement for one removed due to bronze bushing distress and the quantity of bronze debris has not increased, clean and reinstall chip detector, main oil filter and strainers. Return engine to service. Check chip detector and main oil filter after 10 hours and providing no debris (major or minor contaminant) is found at 50 hours, return oil filter and chip detector inspections to operator's standard periodic inspection intervals.

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(c) If the RGB/power section is not a replacement for one removed due to bronze bushing distress and the quantity of bronze debris is the same or has increased, replace the power section within 10 hours of current report of debris.

(16) Laboratory report covering analysis of debris (filter patch or chip detector collected). The amount of individual constituents in oil filter patch debris after analysis is:

(a) Major - when weight of the constituent is more than 50% of the total debris weight.

(b) Minor - when weight of the constituent is less than 50% but more than 5% of the total debris weight.

(c) Traces - when weight of the constituent is less than 5% of total debris weight.

(d) The engine must be replaced if bearing material AMS 6440/6444 or AMS 6490/6491 was found in the debris providing no propeller governor or O/S governor failure , if applicable occurred prior to the generation of debris (i.e., the engine is the source of debris).

(e) The engine may remain in service if bearing material AMS 6440/6444 or AMS 6490/6491 was found in the debris and it was confirmed a propeller governor or O/S governor failure occurred prior to the generation of debris (within 100 hours). A chip detector and main oil filter check must be carried out after 10 hours, and providing no debris is found, after 50 hours. If no debris is found at 50 hours, return oil filter and chip detector to standard periodic inspection intervals. If category 2 or 3 non-allowable debris is found, repeat steps (14) and (15) .

(f) The engine may remain in service if no bearing material (AMS 6444 or AMS 6491) is found and debris is allowable.

(17) Inspect the following components when gear material is found in non-allowable category 2 and 3 debris.

(a) Power Section: First and Second-stage Planet Gears (Ref. 72-00-00, ENGINE, TURBOPROP - INSPECTION).

(b) Gas Generator: Accessory Drive Gears (Ref. 72-00-00, ENGINE, TURBOPROP - INSPECTION).

(c) If debris found during inspection is within limits, engine may remain in service providing a chip detector and main oil filter check is carried out after 10 hours and providing no debris is found, 50 hours. If no debris is found at 50 hours, return oil filter and chip detector to standard periodic inspection intervals. If category 2 or 3 non-allowable debris is found, do the maintenance actions steps (14) or (15) as applicable. If defects found are outside limits, replace the affected module within 10 flight hours.

(18) The engine may remain in service if any other types of category 2 or 3 debris is found as a major or minor filter contaminant. Check chip detector and main oil filter check after 10 hours and providing no debris is found, 50 hours. If no debris is found at 50 hours, return chip detector and oil filter inspection to standard periodic inspection interval. If category 2 or 3 non-allowable debris is found, do the maintenance actions in steps (14) or (15) as applicable.

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NOTE: Air seal material found as a major and minor contaminant when carrying out the first oil filter patch check after installation of a new or overhauled engine is acceptable. However, engine generating seal material (AMS 4117, AMS 4127, AMS 4150, AMS 5613, AMS 5671 or Molybdenum) at a later stage may have had a heavy landing or a defect causing high unbalance (e.g., blade airfoil damage) which can produce seal rub.

C. Patch Making and Chip Detector Debris Transfer

(1) Remove filter element from engine (Ref. Removal/Installation).

(2) Using suitable tapered rubber plugs, seal inner cavity of filter element.

(3) Place filter in cylindrical container.

(4) Fill container with solvent (PWC11-027) or (PWC11-031) to cover filter element leaving a space for liquid movement when shaking. Install cover.

CAUTION: DO NOT USE ULTRASONIC EQUIPMENT.

(5) Shake container for about three minutes manually or 30 seconds in a vibrator. Remove cover.

NOTE: Make sure there are no deposits left on filter. If yes, install container cover and continue shaking.

(6) Use vacuum pump and pour liquid contents of container into the filter holder.

NOTE: Make sure filter holder has new filter paper.

(7) Flush out all possible loose particles which may still be in container by pouring solvent (PWC11-027) or (PWC11-031) into it, shaking, then emptying into filter holder. Remove filter element from container.

(8) Remove filter paper from filter holder and place in receptacle. Identify sample with type of filter, engine number and TSN.

(9) Send sample to laboratory for material spectrographic analysis.

(10) Remove RGB chip detector (Ref. 72-10-00).

(11) Dip magnet end in solvent (PWC11-027) or (PWC11-031) to remove oil.

(12) Transfer debris at magnet end to clear adhesive tape and identify with engine S/N, and TSN or TSO.

(13) Install chip detector (Ref. 72-10-00).

D. Alternative Patch-Making Procedure

CAUTION: DO NOT USE PATCH-MAKING PROCEDURE AS A CLEANING PROCEDURE FOR FILTER ELEMENTS.

(1) Remove filter element from engine (Ref. Removal/Installation).

(2) Using suitable tapered plugs, seal inner cavity of filter element.

(3) Place filter in cylindrical container.

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WARNING: REFER TO MANUFACTURER'S MATERIAL SAFETY DATA SHEETS FOR CONSUMABLE MATERIALS INFORMATION SUCH AS: HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL CHARACTERISTICS, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES.

(4) Fill the container with solvent (PWC11-074) or degreaser (PWC11-084) sufficient to cover the filter element but leave space for liquid movement when shaking and attach the cover.

(5) Shake container for about three minutes manually or 30 seconds in a vibrator. Remove cover.

NOTE: Ensure there are no deposits left on filter. If there is, secure container cover and continue shaking.

(6) Install new filter patch (PWC05-190) in filter patch holder and place over a suitable container (2 liters or ½ gal. min.).

(7) Remove filter element from container.

(8) Pour liquid contents of container into filter patch holder.

(9) Put some solvent (PWC11-074) or degreaser (PWC11-084) into the container to flush out all possible loose particles which may still be in the container. Shake the container and put the liquid into the filter patch holder.

(10) Remove filter patch from filter patch holder and place in patch receptacle. Identify sample with type of filter, engine serial number and hours run.

(11) If any debris is found (particles large enough to be trapped by filter patch), send sample to laboratory for material spectrographic analysis.

NOTE: Some operators retain the filter patches to establish a visual coloration trend. Operators doing this, need to build up their own experience for patch coloration trend evaluation because trends may vary from one engine to another and may be affected by the operating environment and maintenance practices.

(12) Remove plugs from filter element (ensure the preformed packing does not drop inside the element).

WARNING: WHEN USING COMPRESSED AIR FOR DRYING, REGULATE TO 29 psig (200 kPa) OR LESS. WEAR GOGGLES OR FACE SHIELD TO PROTECT EYES.

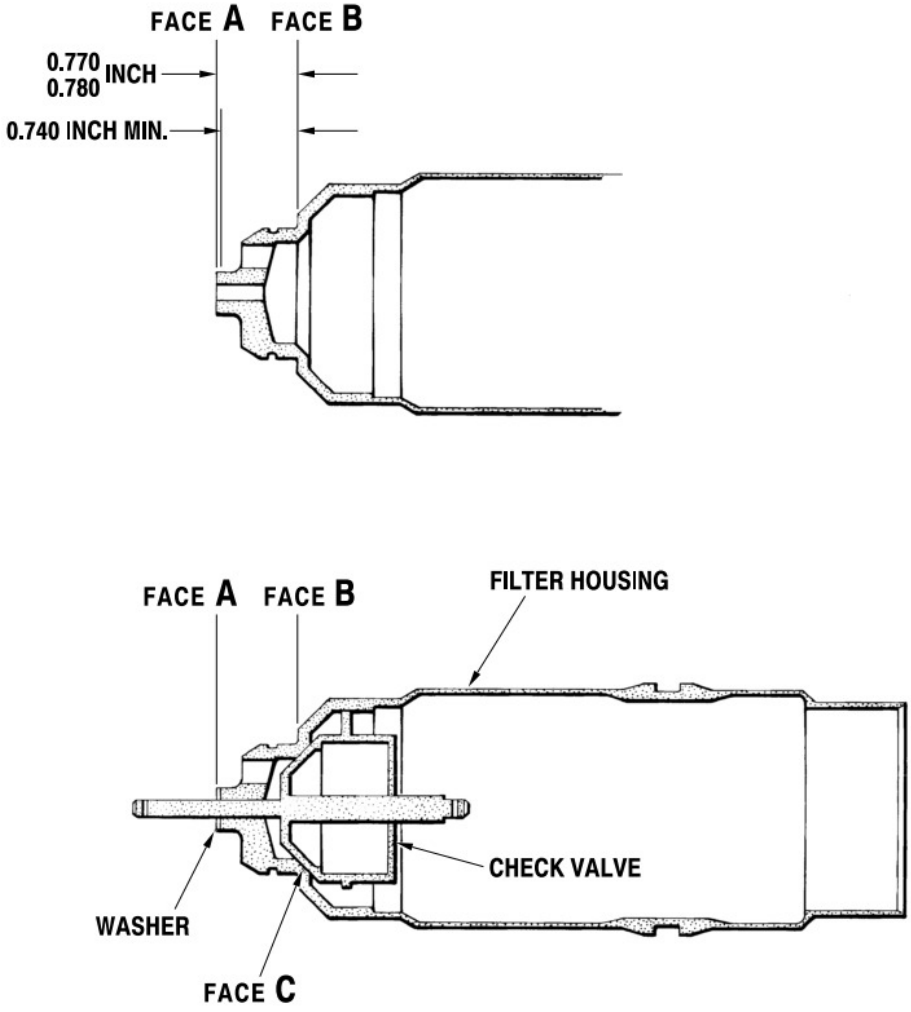
(13) Dry filter element using clean compressed air at 29 psig (200 kPa).

(14) Install filter element on engine (Ref. Removal/Installation).

E. Oil Filter Housing (Pre-SB1247) (Ref. Fig. 205)

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Figure 205 Oil Filter Housing Repair



Pre-SB1247

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(1) If check valve/housing seating is suspect (Ref. Fig. 205, Face C), inspect both seating surfaces for scoring, scratches, nicks, burrs and foreign matter. No damage is acceptable. Damaged seatings should be lapped (Ref. Para. 9.A.).

(2) Inspect check valve spring/filter housing mating face (Ref. Fig. 205, Face A) for pitting, scoring and other damage. Scratches and nicks are acceptable up to 0.005 inch deep. Housings with damage in excess of 0.005 inch deep, must be repaired, and a repair washer installed (Ref. Para. 9.C.).

F. Oil Filter Housing(Post-SB1247)

(1) Inspect check valve preformed packing (14, Fig. 202) for condition by removing the retaining ring (7) and withdrawing the valve assembly from the filter housing. The condition of the packing can be observed by depressing the bypass valve guide (17) and viewing through the windows in the valve housing (13).

(2) No damage is acceptable; replace preformed packing if necessary (Ref. Para. 6.H.).

G. Oil Filter Housing (Post-SB1379)

(1) If check valve/housing seating is suspect, inspect both seating surfaces for scoring, scratches, nicks, burrs and foreign matter. No damage is acceptable. Damaged seatings should be lapped (Ref. Para. 9.B.).

9. Approved Repairs

A. Lapping Oil Filter Check Valve (Pre-SB1247)

(1) Lap face C of the check valve in the filter housing sufficiently to give full 360-degree surface contact; use lapping compound (PWC05-145).

(2) After lapping, thoroughly wash the check valve and filter housing in petroleum solvent (PWC11-027) or (PWC11-031) to remove all traces of lapping compound.

(3) If the check valve is to be installed immediately, reference Para. 6.. If not, store in a suitable polyethylene bag.

B. Lapping Oil Filter Check Valve (Post-SB1379)

(1) If either or both mating surfaces of the check valve and check valve seat are damaged, or if a new part is to be installed, proceed as follows:

(a) Lap the check valve in the check valve seat, sufficiently to give a 0.050 to 0.060 contact band width over the full 360 degree surface contact, using lapping compound (PWC05-145).

(b) After lapping, thoroughly was the check valve and valve seat in petroleum solvent (PWC11-027) or (PWC11-031) to remove all traces of lapping compound.

(2) If the check valve is to be installed immediately, reference Para. 6.H. If not, store in a suitable polyethylene bag.

C. Oil Filter Housing Repair (Pre-SB1247) (Ref. Fig. 205)

(1) In the event of damage to the front face of the housing (Ref. Fig. 205, Face A), proceed as

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follows:

(a) Machine face A, parallel to face B, to a minimum of 0.750 inch from face B.

NOTE: If damage is still evident below this minimum dimension, reject the housing and install a new filter housing.

(b) Apply alodine touch-up solution (PWC05-064) to the machined area.

(c) In order to restore the original dimensions, face A to face B, of 0.770 to 0.780 inch, a repair washer (22, Fig. 201) must be fitted during the installation of the check valve.

(d) At the completion of the repair, tag the filter housing to specify that the fitting of a repair washer is required at installation.

D. Static Leak Test, Oil Filter Check Valve

(1) Install the oil filter housing in the compressor inlet case (Ref. Para. 6.E.).

(2) Fill the oil tank (Ref. 72-00-00, SERVICING).

(3) Do the static leak test:

(a) Let the engine stand for two hours. Monitor the oil leakage into the oil filter housing; the maximum leakage permitted is 0.5 cc/hour.

(b) If the leakage exceeds the limit, verify the condition of the following preformed packings:

1 Pre-SB1247, item 9, Fig. 201.

2 Post-SB1247/Pre-SB1379, item 14, Fig. 202.

3 Post-SB1247/Post-SB1379, item 8 and/or 12, Fig. 202.

(4) If necessary, replace the preformed packing(s) and repeat step (3) .

(5) If the leakage persists, replace part(s) as necessary and/or repeat the appropriate lapping procedure (Ref. Para. A.) or (Ref. Para. B.).

(6) Install the oil filter element into the housing (Ref. Para. 6.E.).

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